

PLUMBOTECTONICS — THE MODEL

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ABSTRACT

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Plumbotectonics is an attempt to model the geochemical behaviour of U, Th and Pb, among major terrestrial reservoirs in agreement with observational data. By recycling rock through the orogenic environment, a dynamically communicating upper crust, lower crust, and mantle can produce the required patterns of lead-isotope evolution.

INTRODUCTION

In several earlier papers (Stacey et al., 1977; Jones et al., 1977; Doe, 1978; Doe and Zartman, 1979; Zartman et al., 1979) we have referred to a model of lead-isotope evolution and made use of growth curves for various geologic environments calculated from it. Our approach to modelling the geologic processes of a long-term, chemically evolving Earth has been named “plumbotectonics” in order to emphasize the dynamic interaction envisioned to take place among crustal and mantle reservoirs. The obvious analogy with plate tectonics recognizes in that concept a mechanism for continuous communication between the diverse crustal and mantle systems. Although the theoretical background of plumbotectonics and a detailed comparison with observed rock and ore data were given by Doe and Zartman (1979), no formal mathematical derivation of the model that generated the lead growth curves has previously been published. The purpose of the present paper, therefore, is to provide such a mathematical formulation and to reexamine some of the numerical parameters in light of current rock and ore data. Although the model first will be derived in close adherence to the original parametric values of Doe and Zartman (1979), an alternate version is also put forth that seeks to reconcile some discrepancies between the original version and more recent estimates of present-day isotopic composition.

The transfer of matter among terrestrial reservoirs is treated as proceeding through cycles — called orogenies — of discrete, sequential operations that are expressed by a series of algebraic equations. From these equations both the total mass distribution and the U, Th, and Pb isotopic contents among reservoirs can be determined as a function of time. For the sake of

clarity, a generalized notational style is employed in deriving the relevant equations, although the transformation of these equations into a form for computer computations has proven to be quite straightforward *.

Most recent attempts to model Pb isotopes (Russell, 1972; Armstrong and Hein, 1973; Zartman, 1974; Cumming and Richards, 1975; Stacey and Kramers, 1975) within gross mantle and crustal reservoirs have recognized isotopic patterns that reflect chemical distinctions among specific geologic environments. But although radioactive decay creates isotopic differences during periods of isolation of one reservoir from another, dynamic processes allowing communications between reservoirs will tend to reduce these differences. Observed isotopic characteristics reflect a dynamic balance between these opposite tendencies and provide important constraints on the extent and timing of differentiation and mixing processes. Lead isotopes are particularly suited to this task because the major crustal and mantle reservoirs contain roughly comparable quantities of U, Th, and Pb and, thus, are *bidirectionally* sensitive to transport among reservoirs. Also, the unique coupling of the $^{238}\text{U} \rightarrow ^{206}\text{Pb}$ and $^{235}\text{U} \rightarrow ^{207}\text{Pb}$ decay schemes offers a time constraint based solely on the lead isotopic compositions and independent of absolute concentration.

BASIC SUPPOSITIONS

The basic suppositions underlying plumbotectonics are largely those of Doe and Zartman (1979), and where modifications have been made they will be noted. A minimum of three broad geologic environments, or reservoirs, having appreciable amounts of U, Th, and Pb and sufficient longevity are required by the model. These reservoirs, the *mantle*, *upper continental crust*, and *lower continental crust*, and an additional short-lived mixing *orogene* provide the necessary chemical environments to produce gross terrestrial isotopic patterns. Hereafter, unless otherwise stated, the term crust will be meant to imply only continental crust. Oceanic crust is not given reservoir status in our model, but its role in contributing to orogene development undoubtedly has considerable importance. In the version of plumbotectonics previously described in Doe and Zartman (1979), the mantle was treated as a single, uniform reservoir that was completely mixed after receiving material back from the orogene. Geochemical and geophysical evidence at the time suggested that only an upper portion of the mantle to a depth of about 500 km actively communicated with the Earth's surface, and our definition of mantle in this restricted sense was accordingly adopted. We proceed here

* The computer program for plumbotectonics written in BASIC language is not presented in this paper, but is separately available as a U.S. Geological Survey Open-file Report [Zartman, R.E. and Doe, B.R., 1980. A Hewlett-Packard 9830A BASIC language program for Plumbotectonics: U.S. Geol. Survey Open-file Report No. 80-1088] that can be obtained from the Open-file Service Section, U.S. Geological Survey, Stop 306, Box 25046, Federal Center, Denver, Colorado, 80225.

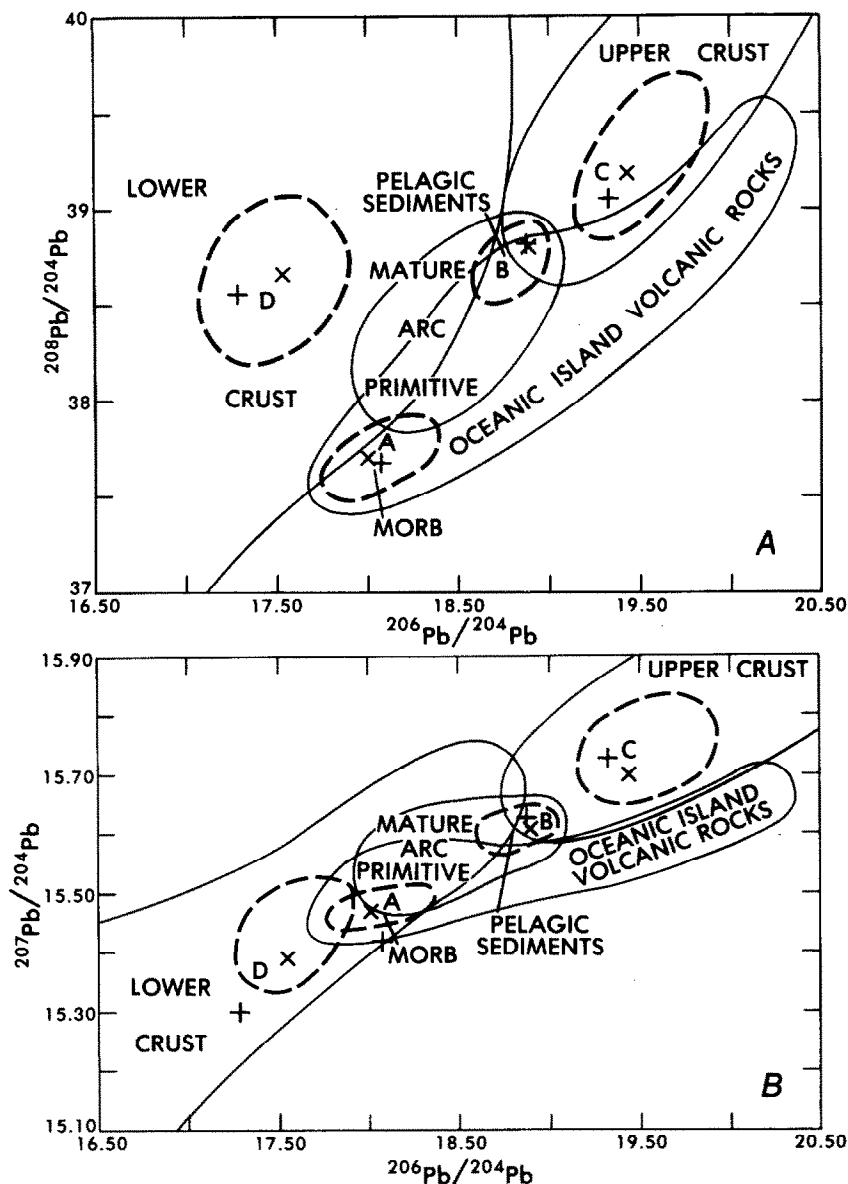


Fig. 1. A, B. Summary plots of lead-isotopic ratios used to define the modern fields for mantle, upper crust, lower crust, and orogene. Solid lines enclose 80% of all data points for each field, and dashed lines enclose probable average values (A = mantle; B = orogene; C = upper crust; and D = lower crust). Some additional geologic environments are also shown. The isotopic composition of modern lead generated by the plumbotectonics model corresponding to each field is shown by an + for version I and by a x for version II; see text for explanation.

with that assumption, but in the final section of this paper we will explore some ramifications of a heterogeneous mantle.

The rock and ore data upon which plumbotectonics is based are stored in the Lead Isotope Data Bank (LIDB) of the U.S. Geological Survey (Doe and Rohrbough, 1977). A summary plot of the lead-isotopic ratios from which we define the modern fields for broad geologic environments is shown in Fig. 1. The upper and lower crust fields are drawn around ~80% of all data points for Mesozoic—Cenozoic rocks and ore deposits and for the present-day isotopic composition of rocks regardless of age that are assigned to these two geologic environments. Inclusion of all analyses would significantly expand the fields and obscure somewhat the essential distinctness of the fields by incorporating a few outlying anomalous ratios. This same reasoning is used to construct the other fields—mature arc, primitive arc, oceanic island volcanics, pelagic sediments, and mid-ocean ridge basalts (MORB)—although here the fields would expand less by including all data points.

Because the model seeks to simulate average lead-isotope evolution curves, we have further attempted to delimit a more restricted field for *average* modern mantle, upper crustal, lower crustal, and orogene lead. Such weighing of the isotopic data is difficult and dependent on a worldwide extrapolation from a very limited amount of known rock composition and age. Nonetheless, a reasonable assessment of the problem leads to values lying within the appropriately labelled dashed areas of Fig. 1. We will return to the matter of selecting model parameters which generate lead isotope evolution curves compatible with these dashed areas later in the paper.

A conspicuous feature shown from the isotopic characterization of mantle, upper crustal, lower crustal, and orogene lead as portrayed in Fig. 1 is the relatively small dispersion in *average* ratios among the various modern fields. Compared with primordial isotopic composition, the spread in $^{206}\text{Pb}/^{204}\text{Pb}$ and $^{208}\text{Pb}/^{204}\text{Pb}$ is about 20 percent, which is commensurate with that of $^{143}\text{Nd}/^{144}\text{Nd}$ and an order of magnitude less than that of $^{87}\text{Sr}/^{86}\text{Sr}$ among equivalent reservoirs. While the isotopic variation recorded in the Sm-Nd and Rb-Sr systematics mainly reflects chemical fractionation between the crust and mantle, lead isotopes cannot be so simply explained. Instead, both the most and the least radiogenic lead occur in the continental crust attesting to an unequal partitioning of U, Th, and Pb in the upper and lower portions of the crust. The intermediate position on the $^{206}\text{Pb}/^{204}\text{Pb}$ vs. $^{207}\text{Pb}/^{204}\text{Pb}$ plot of the MORB field plays a key role in our interpretation that little fractionation of U/Pb takes place between the total crust and the mantle. Nevertheless, if, as we contend, MORB does represent the dominant mantle contribution to the orogene, then its source reservoir can be regarded as complementary in U/Pb to the total crust even though strong parent-daughter fractionation is absent. A more complex lead-isotopic pattern is found in some oceanic island volcanic rocks, but this lead will be explained in a later section of the paper as most likely having a secondary origin.

Controversy continues over the nature and timing of the primary chemical

differentiation of the Earth and subsequent maturation of the continental crust. For our purposes we need make only very general assumptions in this regard, although, as will be seen, the timing of crustal separation from the mantle has an important bearing on the relative placement of isotopic fields in Fig. 1. Without entering the debate over the age of the Earth or the nature of the earliest planetary differentiation, we simply use the empirical data of Stacey and Kramers (1975) to constrain our growth curve for orogene lead. More pertinent here is the accumulating evidence from the oldest crustal rocks that sometime around 4 b.y. ago an accretional stage of continental growth began. Thereafter, further evolution of the continental crust involved recurrent orogenies that mixed together some fraction of the mantle and portions of preexisting crust to form new segments of crust. Just how regular continental growth has been is not well known, although for simplicity we have assumed it to be constant in magnitude over time. Our model tolerates moderate deviation from this assumption, and still maintains the relative position of lead isotope fields for the different environments.

Unlike the Rb-Sr and Sm-Nd systems, which to a first order only require consideration of a gross crust and mantle differentiation to explain their isotopic evolution, the U-Th-Pb system clearly records an important intracrustal partitioning affecting the element ratios. The dichotomous relationship between upper and lower continental crust is made quite obvious by the almost mutually exclusive fields of these two environments as shown in Fig. 1. Yet despite the separation of fields, which generally becomes greater with increasing age of the rock, lead of a given age throughout the entire crust is often found to be similar in its calculated "initial" isotopic composition. Indeed, even since the earliest mass spectrometric analyses, this "initial" lead incorporated into rocks and ores was recognized as following a remarkably uniform isotopic growth with time. Although examples of complex evolution in the radiogenic isotopes were also discovered, such lead was regarded as anomalous compared to that with the more regular pattern, which seemed to dominate the continental lead budget. An assortment of models, in which U/Pb either remains constant or varies only slightly with time, has been proposed to describe the source reservoir of this so-called normal lead.

The dichotomy between the upper and lower crust, then, is seen to result from different rates of isotopic evolution occurring after rather than before the present host rock of the lead formed. Furthermore, the observed pattern in modern crustal lead can be readily explained in terms of the unique geochemical behavior of U, Th, and Pb in the crust. We now know that the upper crust, on the average, has a $^{238}\text{U}/^{204}\text{Pb}$ considerably higher than predicted by a closed-system model, and, consequently, this high parent-daughter ratio produces by in situ decay a rather radiogenic lead with the passage of time. Attempts to calculate an average isotopic composition for the present-day upper crust is fraught with uncertainty, but must be the most enriched in radiogenic isotopes of the major reservoirs under consideration.

The lower portion of established crust is frequently composed of pyroxene granulite or other highly metamorphosed rocks that have quite low uranium contents and $^{238}\text{U}/^{204}\text{Pb}$. Thorium and lead contents do not appear to be so drastically reduced from upper crustal values; the $^{232}\text{Th}/^{204}\text{Pb}$ is often highly variable and may average only slightly less than its upper crustal value. This preferential depletion of uranium appears usually to arise during major orogenic episodes associated with formation of new segments of continental crust. The accumulation of lead in the lower crust relatively unsupplemented by subsequent radioactive decay has a significant impact on future lead-isotope systematics, and the complementary excess of uranium in the upper crust can explain the opposite effect seen there.

Primarily, three factors will be considered as influencing the geochemical behavior of U, Th, and Pb among the terrestrial reservoirs; they are: (1) element partitioning ratios; (2) time of crust formation; and (3) the degree to which preexisting chemical differences are homogenized by recycling. An attempt to bring these three factors into balance with the observational data is really the reason that the plumbotectonics model was derived.

MATHEMATICAL DERIVATION

We now proceed to express the plumbotectonic model by general algebraic equations. This derivation of the model can be used directly to generate the reservoir masses, elemental abundances and concentrations, and isotopic compositions at the time of any orogeny. As originally put forth, the model dealt only with the isotopic systematics of uranium, thorium, and lead, but in its general form any radioactive parent and radiogenic daughter system can be treated. To facilitate the reading of this section, Table I summarizes the notation used in the mathematical derivation. Table II gives brief description of the model (version I) as previously presented by Doe and Zartman (1979).

At time, t , at the onset of the j th orogeny, an orogene of mass, M_t^o , is created from increments of the mantle, ΔM_t^m , and of preexisting segments of upper, $\Delta M_t^{u(i)}$, and lower $\Delta M_t^{l(i)}$, continental crust, i.e.:

$$M_t^o = \Delta M_t^m + \sum_i (\Delta M_t^{u(i)} + \Delta M_t^{l(i)}) \quad (1)$$

for all $i < j$ and where $\underline{t}$ refers to reservoir conditions just prior to the orogeny. Throughout this paper we adopt the convention of counting geologic time, t , backward from the present, and of numbering orogenies consecutively from the oldest (first) to the youngest (last). All separate chemical and isotopic identity of the contributing increments is lost by homogenization, and the orogene is immediately redistributed into newly formed segments of upper, $M_t^{u(j)}$, and lower, $M_t^{l(j)}$, continental crust and returned mantle, ΔM_t^m :

$$M_t^o = \Delta M_{\bar{t}}^m + M_{\bar{t}}^{u(j)} + M_{\bar{t}}^{l(j)} \quad (2)$$

Here $\bar{t}$ refers to reservoir conditions immediately after the orogeny.

TABLE I

Notation used in the mathematical derivation of the plumbotectonics model

$M_t^m, M_t^{u(i)}, M_t^{l(i)}$	Mass of mantle, <i>i</i> th segment of upper crust, and <i>i</i> th segment of lower crust, respectively, immediately prior to an orogeny at time <i>t</i>
$M_t^m, M_t^{u(i)}, M_t^{l(i)}$	Mass of mantle, <i>i</i> th segment of upper crust, and <i>i</i> th segment of lower crust, respectively, immediately following an orogeny at time <i>t</i>
M_t^o	Mass of orogene during an orogeny at time <i>t</i>
$\Delta M_t^m, \Delta M_t^{u(i)}, \Delta M_t^{l(i)}$	Incremental mass of mantle, <i>i</i> th segment of upper crust, and <i>i</i> th segment of lower crust, respectively, used to create an orogene at time <i>t</i>
$\Delta M_t^m, M_t^{u(j)}, M_t^{l(j)}$	Incremental mass returned to the mantle, mass of new (<i>j</i>)th segment of upper crust, and mass of a new (<i>j</i>)th segment of lower crust, respectively, representing the redistribution from an orogene at time <i>t</i>
$(\Delta M_t^m / M_t^m)$	Mass fraction of mantle contributing to the creation of an orogene at time <i>t</i>
f_v, f_h	Erosional (vertical) and areal (horizontal) functions, respectively. f_v is the fractional height of preexisting segments of upper crust and f_h is the fractional area of preexisting segments of total crust contributing to the creation of an orogene at time <i>t</i>
$\alpha N_t^m, \alpha N_t^{u(i)}, \alpha N_t^{l(i)}$	Moles of isotopic species α in the mantle, <i>i</i> th segment of upper crust, and <i>i</i> th segment of lower crust, respectively, immediately prior to an orogeny at time <i>t</i>
$\alpha N_t^m, \alpha N_t^{u(i)}, \alpha N_t^{l(i)}$	Moles of isotopic species α in the mantle, <i>i</i> th segment of upper crust, and <i>i</i> th segment of lower crust, respectively, immediately following an orogeny at time <i>t</i>
αN_t^o	Moles of isotopic species α in the orogene during an orogeny at time <i>t</i>
$\Delta \alpha N_t^m, \Delta \alpha N_t^{u(i)}, \Delta \alpha N_t^{l(i)}$	Moles of isotopic species α contained in the incremental additions to the orogene from the mantle, <i>i</i> th segment of upper crust, and <i>i</i> th segment of lower crust, respectively, at time <i>t</i>
$\Delta \alpha N_t^m, \alpha N_t^{u(j)}, \alpha N_t^{l(j)}$	Moles of isotopic species α contained in the increment returned to the mantle, new (<i>j</i>)th segment of upper crust, and new (<i>j</i>)th segment of lower crust, respectively, representing the redistribution from an orogene at time <i>t</i>
$\alpha E^m, \alpha E^u, \alpha E^l$	Enrichment factors for isotope species α (equal to the concentration of the species in the incremental additions relative to its concentration in the reservoir) for the mantle, upper crust, and lower crust, respectively
$\alpha F^m, \alpha F^u, \alpha F^l$	Partitioning ratios for isotopic species α (equal to the concentration ratios of the species among the returning components from an orogene) for the mantle, upper crust, and lower crust, respectively. $\alpha F^m + \alpha F^u + \alpha F^l = 1$

TABLE II

Parameters of the plumbotectonics lead isotope evolution model (version I)

- I. Starting conditions (all material in undifferentiated upper mantle to 500 km depth [800×10^{24} g] at 4.0 b.y. ago):
- A. Lead isotopic composition: $^{206}\text{Pb}/^{204}\text{Pb} = 10.36$; $^{207}\text{Pb}/^{204}\text{Pb} = 12.12$; $^{208}\text{Pb}/^{204}\text{Pb} = 30.55$
- B. Element abundances: $^{204}\text{Pb} = 38.0 \times 10^{15}$ moles; $^{238}\text{U} = 349 \times 10^{15}$ moles; $^{232}\text{Th} = 1335 \times 10^{15}$ moles
- II. Eleven orogenies evenly spaced in time; i.e., at 4.0 b.y., 3.6 b.y., 3.2 b.y., ..., 0.0 b.y. During each orogeny:
- A. Of the material entering the orogene, (1) the mantle contributes 1/8 of itself, at 4.0 b.y., 1/16 at 3.6 b.y., 1/32 at 3.2 b.y., 1/64 at 2.8 b.y., and 1/128 thereafter; (2) the upper crust contributes 3/10 of remaining amounts of each older segment (erosional function); and (3) the total crust contributes 1/10 of remaining amounts of each older segment (areal function). Components (2) and (3) do not pertain to the first orogeny.
- B. The contents of the orogene are chemically and isotopically homogenized.
- C. 2.6×10^{24} g of new upper crust and 2.6×10^{24} g of new lower crust are created, and remaining material returns from the orogene to the mantle. Lead, uranium, and thorium distribute themselves according to the following partitioning ratios:

Element	Mantle	Upper crust	Lower crust
Lead	0.028	0.754	0.218
Uranium	0.024	0.854	0.122
Thorium	0.020	0.788	0.192

III. Ending conditions (prevailing today at time of final orogeny):

A. Lead isotopic composition:

Environment	$\frac{^{206}\text{Pb}}{^{204}\text{Pb}}$	$\frac{^{207}\text{Pb}}{^{204}\text{Pb}}$	$\frac{^{208}\text{Pb}}{^{204}\text{Pb}}$	$\frac{^{238}\text{U}}{^{204}\text{Pb}}$	$\frac{^{232}\text{Th}}{^{238}\text{U}}$
Mantle	18.08	15.42	37.68	8.90	3.58
Oroge	18.88	15.63	38.82	10.88	3.62
Upper crust contributed to orogene	19.33	15.73	39.06	12.22	3.40
Lower crust contributed to orogene	17.29	15.30	38.56	5.90	5.94

B. Element abundances and concentrations:

Reservoir	Mass ($\times 10^{24}$ g)	Abundances ($\times 10^{15}$ moles)			Average concentrations (ppm)		
		^{204}Pb	^{238}U	^{232}Th	Pb	U	Th
Mantle	775.2	19.5	174	619	0.39	0.054	0.19
Upper crust	7.0	10.3	127	430	23.5	4.4	14.3
Lower crust	17.8	8.2	48	286	6.8	0.64	3.7

NOTE: Modifications in the model equations (see text) and minor calculation errors in the original treatment cause this table to differ slightly from the corresponding table of Doe and Zartman (1979). The results are in close agreement and represent essentially the same version of the plumbotectonics model.

Numerically evaluating the model parameters pertaining to the creation of the orogene as given in Table II, we can define the various increments put into the orogene at each orogeny as follows:

$$\Delta M_{\underline{t}}^m = (\Delta M_{\underline{t}}^m / M_{\underline{t}}^m) \cdot M_{\underline{t}}^m \quad (3)$$

where $(\Delta M_{\underline{t}}^m / M_{\underline{t}}^m) = 1/8$ at 4.0 b.y., $1/16$ at 3.6 b.y., $1/32$ at 3.2 b.y., $1/64$ at 2.8 b.y., and $1/128$ thereafter. Also, because the contribution from each segment of preexisting upper crust is both a fraction of its height, f_v , and a fraction of its area, f_h , but the contribution from each segment of preexisting lower crust is only a fraction of its area, f_h , we have:

$$M_{\underline{t}}^{u(i)} = M_{\underline{t}}^{u(i)}(1 - f_v)(1 - f_h)$$

and

$$M_{\underline{t}}^{l(i)} = M_{\underline{t}}^{l(i)}(1 - f_h)$$

which gives us the relationships:

$$\Delta M_{\underline{t}}^{u(i)} = M_{\underline{t}}^{u(i)} - M_{\underline{t}}^{u(i)} = M_{\underline{t}}^{u(i)}(f_v + f_h - f_v \cdot f_h) \quad (4)$$

and

$$\Delta M_{\underline{t}}^{l(i)} = M_{\underline{t}}^{l(i)} - M_{\underline{t}}^{l(i)} = M_{\underline{t}}^{l(i)} \cdot f_h \quad (5)$$

Doe and Zartman (1979) refer to f_v and f_h as the erosional (vertical) and areal (horizontal) functions, respectively, and show that they are equivalent to an exponential erosion of the upper crust and consumption of the total crust. Good agreement between model and Earth is achieved if the height of older upper crustal segments is reduced by $3/10$ and the area of older crust is reduced by $1/10$ of their preceding values during every subsequent orogeny. Evaluating equations (4) and (5) for $f_v = 3/10$ and $f_h = 1/10$ we have:

$$\Delta M_{\underline{t}}^{u(i)} = 0.37 M_{\underline{t}}^{u(i)} \quad (4')$$

and

$$\Delta M_{\underline{t}}^{l(i)} = 0.10 M_{\underline{t}}^{l(i)} \quad (5')$$

Following each orogeny the mantle can be described by:

$$M_{\underline{t}}^m = M_{\underline{t}}^m - \Delta M_{\underline{t}}^m + \Delta M_{\underline{t}}^m \quad (6)$$

Preexisting segments of upper and lower continental crust become:

$$M_{\underline{t}}^{u(i)} = M_{\underline{t}}^{u(i)} - \Delta M_{\underline{t}}^{u(i)} \quad (7)$$

and

$$M_{\underline{t}}^{l(i)} = M_{\underline{t}}^{l(i)} - \Delta M_{\underline{t}}^{l(i)} \quad (8)$$

or, substituting from eqs. 4' and 5':

$$M_{\underline{t}}^{u(i)} = 0.63 M_{\underline{t}}^{u(i)} \quad (7')$$

and

$$M_{\bar{t}}^{1(i)} = 0.90M_{\bar{t}}^{1(i)} \quad (8')$$

The mass of the newly formed segment of upper and lower continental crust is defined in Table II as:

$$M_{\bar{t}}^{u(j)} = M_{\bar{t}}^{l(j)} = 2.6 \times 10^{24} \text{ g} \quad (9)$$

Furthermore, from the initial boundary conditions stated in Table II, we can see that at the time of the first orogeny ($j = 1$) 4.0 b.y. ago $M_{\bar{t}}^m = 800 \times 10^{24} \text{ g}$ and $M_{\bar{t}}^{u(i)} = M_{\bar{t}}^{l(i)} = 0$ for all values of i ; i.e., all material resides in undifferentiated upper mantle and no continental crust yet exists. Our model for the transfer of total mass among the three groupings of reservoirs designated mantle, upper continental crust, and lower continental crust is now completely defined, and it may be cycled through the 11 evenly spaced orogenies occurring between 4.0 b.y. ago and the present. The final configuration of mantle, upper crust, and lower crust is shown in Fig. 2, and the mass distribution for the various groupings of reservoirs as a function of time is given in Table III.

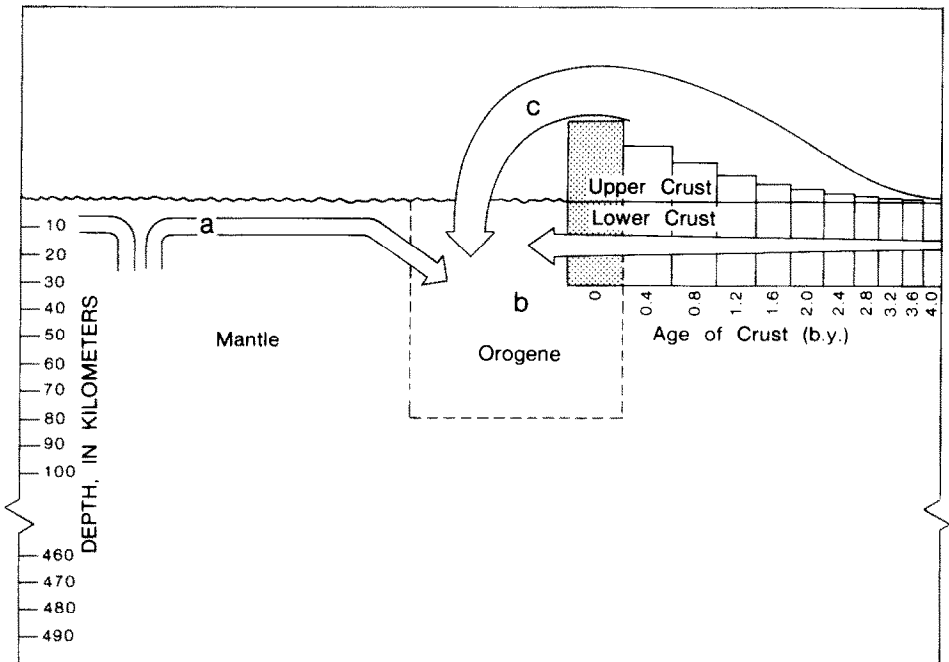


Fig. 2. Final configuration of reservoirs showing apportionment of mantle, upper crust, lower crust, and orogene. The arrows represent the transport of material into the orogene during the last orogenic cycle, which, in turn, produced the stippled block of new crust. Specific environments discussed in text are a, mantle, b, orogene, c, upper crust contributed to orogene, and d, lower crust contributed to orogene.

TABLE III

Mass distribution for the various groupings of reservoirs as a function of time.

$M \frac{u(1)}{4.0}$ and $M \frac{l(1)}{4.0}$ are also given to illustrate the decrease in size of crustal reservoirs with successive orogenies. [All masses in units of 10^{24} g]

i	t (b.y.)	$M \frac{m}{t}$	$\Sigma M \frac{u(i)}{t}$	$\Sigma M \frac{l(i)}{t}$	$M \frac{u(1)}{4.0}$	$M \frac{l(1)}{4.0}$
0	>4.0	800.0	0.000	0.000	0.000	0.000
1	4.0	794.8	2.600	2.600	2.600	2.600
2	3.6	790.8	4.238	4.940	1.638	2.340
3	3.2	787.7	5.270	7.046	1.032	2.106
4	2.8	785.1	5.920	8.941	0.650	1.895
5	2.4	783.0	6.330	10.647	0.410	1.706
6	2.0	781.2	6.588	12.183	0.258	1.535
7	1.6	779.7	6.750	13.564	0.163	1.382
8	1.2	778.3	6.853	14.808	0.102	1.244
9	0.8	777.1	6.917	15.927	0.065	1.119
10	0.4	776.1	6.958	16.934	0.041	1.007
11	0.0	775.2	6.983	17.841	0.026	0.907

Let us go on to determine the distribution of the pertinent isotopic species among the model reservoirs as a function of geologic time. We define ${}^{\alpha}N_t^m$, ${}^{\alpha}N_t^{u(i)}$, and ${}^{\alpha}N_t^{l(i)}$ as the number of moles of isotopic species α in the mantle, upper continental crust, and lower continental crust immediately preceding an orogeny, and ${}^{\alpha}N_t^m$, ${}^{\alpha}N_t^{u(i)}$ and ${}^{\alpha}N_t^{l(i)}$ as the same parameters immediately after the orogeny. Likewise, $\Delta^{\alpha}N_t^m$, $\Delta^{\alpha}N_t^{u(i)}$, and $\Delta^{\alpha}N_t^{l(i)}$ are the number of moles of species α contained in the incremental additions to the orogene from the mantle, upper continental crust, and lower continental crust, respectively. A corresponding term for the number of moles of species α returned from the orogene needs only be defined for the mantle, $\Delta^{\alpha}N_t^m$. Segments of preexisting crust are partially destroyed during an orogeny but do not receive material back from the orogene in our model. For the newly formed j th segment of crust being generated entirely from material of the orogene, we define ${}^{\alpha}N_t^{u(j)}$ and ${}^{\alpha}N_t^{l(j)}$ as the number of moles of species α acquired by the upper and lower continental crust, respectively.

From material balance considerations for the orogene where ${}^{\alpha}N_t^o$ is the number of moles of species α in the orogene we have:

$${}^{\alpha}N_t^o = \Delta^{\alpha}N_t^m + \sum_i (\Delta^{\alpha}N_t^{u(i)} + \Delta^{\alpha}N_t^{l(i)}) \quad (10a)$$

$$= \Delta^{\alpha}N_t^m + {}^{\alpha}N_t^{u(j)} + {}^{\alpha}N_t^{l(j)} \quad (10b)$$

The net change in the number of moles of species α experienced by each

reservoir due to the orogeny is:

$$\alpha N_T^m = \alpha N_T^m - \Delta \alpha N_T^m + \Delta \alpha N_T^m \quad (11)$$

$$\alpha N_T^{u(i)} = \alpha N_T^{u(i)} - \Delta \alpha N_T^{u(i)} \quad (12)$$

$$\alpha N_T^{l(i)} = \alpha N_T^{l(i)} - \Delta \alpha N_T^{l(i)} \quad (13)$$

for all $i < j$.

We can define the behavior of each isotopic species in the context of total mass transfer among the reservoirs. In the original model of Doe and Zartman (1979) no fractionation of chemical species was considered between an overall reservoir and the increment derived therefrom used to create the orogene. Instead, they related the amount of mantle involvement to the volume of extracted source material rather than to the volume of resultant volcanic rock. Because, in general, different chemical species do experience different degrees of fractionation during partial melting, the contribution of the mantle to the orogene will here be expressed as a mass ratio of magma to total mantle to a depth of 500 km. Thus, this ratio is revised from that used by Doe and Zartman (1979) to values of 1/8 at 4.0 b.y., 1/16 at 3.6 b.y., 1/32 at 3.2 b.y., 1/64 at 2.8 b.y., and 1/128 thereafter. To compensate for this revision we introduce enrichment factors into the model equations so that the numerical solution given in this paper agrees with our earlier treatment. The following equations are written to describe the incremental additions of species α to the orogene:

$$\Delta \alpha N_T^m = (\Delta M_T^m / M_T^m) \cdot \alpha N_T^m \cdot \alpha E^m \quad (14)$$

$$\Delta \alpha N_T^{u(i)} = (\Delta M_T^{u(i)} / M_T^{u(i)}) \cdot \alpha N_T^{u(i)} \cdot \alpha E^u \quad (15)$$

$$\Delta \alpha N_T^{l(i)} = (\Delta M_T^{l(i)} / M_T^{l(i)}) \cdot \alpha N_T^{l(i)} \cdot \alpha E^l \quad (16)$$

Here αE^m , αE^u , and αE^l are the enrichment factors for species α (equal to the concentration of the species in the increment relative to its concentration in the reservoir) for the mantle, upper crust, and lower crust, respectively. At the scale of our model, we would expect the enrichment factors to be essentially equal for isotopic species of the same chemical element, and, as a simplifying assumption, we will also here consider the factors applied to each reservoir to be equal for lead, uranium, and thorium. This assumption does not mean that differences in geochemical behaviors among these elements are entirely ignored, however, and the important role of chemical fractionation will be accommodated by the use of partitioning ratios at the completion of the orogenic cycle. By setting $\alpha E^m = 4$ in order to simulate complete extraction of Pb, U, and Th into a 25% melt representing average (i.e., MORB-dominated) mantle contribution to the orogene, we maintain consistency with the original model. No need is seen for a similar fractionation bias for the upper and lower crustal increments, which, at least grossly, enter the orogene by total removal of large volumes from the contributing reservoirs. Accordingly, we set $\alpha E^u = \alpha E^l = 1$ throughout the rest of this paper.

Equations 14, 15 and 16 can now be combined with eqs. 3, 4' and 5' for a numerical evaluation of the incremental content relative to the reservoir content of each isotopic species.

The redistribution of material from the orogene back into mantle and newly formed upper and lower crustal reservoirs may also be described. However, chemical fractionation does now take place among the returning increments, and this feature leads to the isotopic distinctions among reservoirs so critical to the model. We treat this fractionation in terms of partitioning ratios — ${}^{\alpha}F^m$, ${}^{\alpha}F^u$, and ${}^{\alpha}F^l$ for the mantle, upper crust, and lower crust, respectively — and defined as the concentration ratios for species among the returning increments. Although the partitioning ratios have the same values for all isotopes of the same chemical element, they are different for Pb, U, and Th as given in Table II. As was done with the introduction of enrichment factors in creating the orogene, the partitioning ratios have also been adjusted from the values of Doe and Zartman (1979) to compensate for the revised treatment of mantle contribution to the orogene. These ratios, of course, pertain to chemical distribution among the gross reservoirs rather than to any individual local mineral assemblage. The weighted redistribution from the orogene can readily be given as:

$$\Delta {}^{\alpha}N_t^m = {}^{\alpha}N_t^o (\Delta M_t^m \cdot {}^{\alpha}F^m / s) \quad (17)$$

$${}^{\alpha}N_t^{u(j)} = {}^{\alpha}N_t^o (M_t^{u(j)} \cdot {}^{\alpha}F^u / s) \quad (18)$$

$${}^{\alpha}N_t^{l(j)} = {}^{\alpha}N_t^o (M_t^{l(j)} \cdot {}^{\alpha}F^l / s) \quad (19)$$

where:

$$s = \Delta M_t^m \cdot {}^{\alpha}F^m + M_t^{u(j)} \cdot {}^{\alpha}F^u + M_t^{l(j)} \cdot {}^{\alpha}F^l$$

Equations 17, 18 and 19 can now be combined with eqs. 2 and 9 for a numerical evaluation in terms of orogene mass and content of each species. M_t^o and ${}^{\alpha}N_t^o$ can, in turn, be functionally related back to the reservoirs contributing to the creation of the orogene from eqs. 1, 3, 4' and 5' and eqs. 10a, 14, 15 and 16, respectively. Thus, by knowing the mass and content of the reservoirs just prior to an orogeny we can calculate the mass and content of them immediately after an orogeny. Because no chemical or isotopic transport takes place among reservoirs except through an orogene, the model is now defined at any time given initial boundary conditions for isotopic species *not affected by radioactive decay*.

The accumulation of radiogenic species from the decay of their radioactive parents during residence in reservoirs following one orogeny and preceding the next orogeny remains to be considered. Orogenies themselves are modelled as instantaneous events and we need not treat radioactive decay in an orogene. By convention, we give all equations in terms of the parent isotope normalized to its equivalent modern abundance; i.e., corrected for radioactive decay. Thus, we need only to account for the increase in abundances of daughter isotopes acquired between two successive orogenies occurring at times t'

(older) and t'' (younger). If we call species α the radioactive parent, species β its radiogenic daughter, and λ the corresponding decay constant, we have:

$$\beta N_{t'}^m = \beta N_{t'}^m + \alpha N_{t'}^m (e^{\lambda t'} - e^{\lambda t''}) \quad (20)$$

$$\beta N_{t'}^{u(i)} = \beta N_{t'}^{u(i)} + \alpha N_{t'}^{u(i)} (e^{\lambda t'} - e^{\lambda t''}) \quad (21)$$

$$\beta N_{t'}^{l(i)} = \beta N_{t'}^{l(i)} + \alpha N_{t'}^{l(i)} (e^{\lambda t'} - e^{\lambda t''}) \quad (22)$$

This set of equations completes the mathematical formulation of the model, which can now be used to describe the isotopic evolution of a radiogenic species within a recycling mantle—upper crust—lower crust system. Some of the significant isotope composition and concentration values are presented in Table IV and Figs. 3 and 4. The specific version (version I) of the model derived here with the parameters of Table II is but one of a family

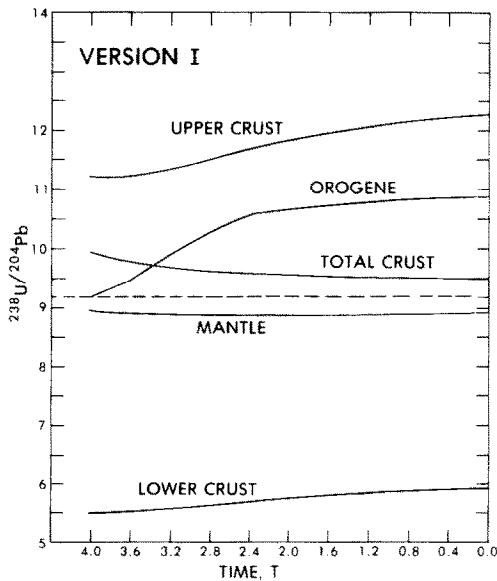


Fig. 3. $^{238}\text{U}/^{204}\text{Pb}$ generated by the plumbotectonics model (version I) for the mantle, orogene, and newly formed segments of upper crust, lower crust, and total crust as a result of each orogeny. The horizontal dashed line represents the entire system, which has a constant value of 9.18.

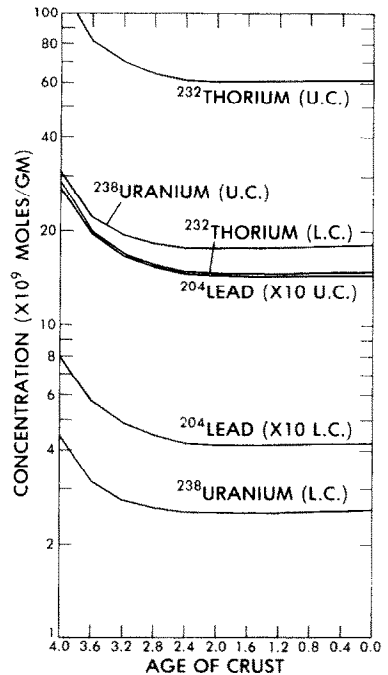


Fig. 4. ^{204}Pb , ^{238}U , and ^{232}Th concentrations in individual segments of upper (u.c.) and lower crust (l.c.) generated by the plumbotectonics model (version I). The higher values of the first several orogenies reflect the early, rapid differentiation of the mantle given by this version of the model; otherwise, there is no significant change in concentration with time.

TABLE IV

Lead-isotopic composition for reservoir growth curves generated by the plumbotectonics model (version I)

t (b.y.)	Mantle				Orogene				Upper crust				Lower crust			
	$\frac{206\text{Pb}}{204\text{Pb}}$	$\frac{207\text{Pb}}{204\text{Pb}}$	$\frac{208\text{Pb}}{204\text{Pb}}$		$\frac{206\text{Pb}}{204\text{Pb}}$	$\frac{207\text{Pb}}{204\text{Pb}}$	$\frac{208\text{Pb}}{204\text{Pb}}$		$\frac{206\text{Pb}}{204\text{Pb}}$	$\frac{207\text{Pb}}{204\text{Pb}}$	$\frac{208\text{Pb}}{204\text{Pb}}$		$\frac{206\text{Pb}}{204\text{Pb}}$	$\frac{207\text{Pb}}{204\text{Pb}}$	$\frac{208\text{Pb}}{204\text{Pb}}$	
4.0	10.36	12.12	30.55		10.36	12.12	30.55		10.36	12.12	30.55		10.36	12.12	30.55	
3.6	11.36	13.21	31.34		11.42	13.27	31.40		11.61	13.48	31.57		10.98	12.79	31.41	
3.2	12.30	13.94	32.11		12.47	14.09	32.30		12.69	14.29	32.46		11.76	13.46	32.24	
2.8	13.19	14.44	32.86		13.50	14.67	33.20		13.72	14.83	33.35		12.52	13.95	33.06	
2.4	14.02	14.78	33.59		14.47	15.06	34.11		14.70	15.20	34.23		13.27	14.33	33.89	
2.0	14.80	15.00	34.30		15.35	15.29	34.95		15.64	15.44	35.10		14.00	14.61	34.70	
1.6	15.54	15.15	35.00		16.17	15.44	35.77		16.51	15.59	35.94		14.72	14.83	35.51	
1.2	16.23	15.26	35.69		16.92	15.53	36.56		17.30	15.67	36.76		15.40	15.00	36.29	
0.8	16.88	15.33	36.37		17.62	15.58	37.34		18.04	15.71	37.55		16.06	15.13	37.06	
0.4	17.50	15.38	37.03		18.27	15.61	38.09		18.71	15.73	38.32		16.69	15.22	37.82	
0.0	18.08	15.42	37.68		18.88	15.63	38.82		19.33	15.73	39.06		17.29	15.30	38.56	

of solutions. Other versions may be produced by varying certain of the model parameters, such as the mass and species abundance of the involved mantle, the mass of the newly created crust, $(\Delta M_t^m/M_t^m)$ as a function of time, f_v , f_h , ${}^\alpha E$, ${}^\alpha F$, and so forth.

Useful terms relating to any transport model of this type are: (1) the mean age of the mass of the crust; and (2) the mean age of species α in the crust (Jacobsen and Wasserburg, 1979). We need not elaborate on the derivation of these terms, which are given in general form by Jacobsen and Wasserburg (1979). We do note that in addition to the effect of a fractionating mantle considered by those authors, our model also allows for the return of crustal mass and isotopic species to the mantle. The mean age takes into account only material that has survived in the crust until today. Furthermore, if matter enters the crust and is returned to the mantle, and later reenters the crust, we ignore its "first residence" and consider only its "second residence" in calculating mean age.

Because the redistribution of total mass out of the orogene is not vigorously defined, we must make some assumption about the fate of older crustal segments contributed to the orogene. According to our model, preexisting crust that passes through an orogeny is partially used to create new crust and partially returns to the mantle. In calculating the mean age of the mass of the crust, the former fraction should be accorded its original age of entry into the crust and the latter fraction is now lost from the crustal budget. As one limit, all recycled older crust would go to form new crust, which will lead to a maximum mean age, $\bar{T}_{M,\max}$; as the other limit, all recycled older crust would return to the mantle, which will lead to a minimum mean age, $\bar{T}_{M,\min}$. This redistribution is defined for individual species α , but every chemical element would have to be considered to specify total mass. Written in our notational style, we have:

$$\bar{T}_{M,\max} = \frac{\sum_t (M_t^m - M_T^m) \cdot t}{\sum_t (M_t^m - M_T^m)} \quad (23)$$

where the summation is for all t . Likewise, we have:

$$\bar{T}_{M,\min} = \frac{\sum_i (M_T^{u(i)} + M_T^{l(i)}) \cdot \tau}{\sum_i (M_T^{u(i)} + M_T^{l(i)})} \quad (24)$$

where the summation for $i = 1$ to 11 is made for the final configuration of reservoirs, and τ [= 0.4(11 - i)] is the age of each crustal segment. For version I — as well as version II to be derived in the following section — of the

plumbotectonics model $\bar{T}_{M,\max} = 2.67$ b.y. and $\bar{T}_{M,\min} = 1.33$ b.y. Neither of these limiting values is strictly realistic, but $\bar{T}_{M,\max}$ is probably a reasonable approximation when we consider the difficulty in subducting large quantities of preexisting crust back into the mantle.

Because the partitioning ratios have been defined for Pb, U, and Th, we can calculate a precise mean age in the crust for these specific elements. Furthermore, an element preferentially enriched in crustal reservoirs relative to the returned mantle upon redistribution from the orogene can have a $\bar{T}_\alpha > \bar{T}_{M,\max}$. In order to calculate $\bar{T}_\alpha$ we must take inventory of all crustal segments and determine when each atom actually entered the crust from the mantle. An atom may have been recycled several times into progressively younger crustal segments, but we are only interested in the time of initial appearance in the crust when computing the mean age. This inventory can be made for atoms of age t by modifying eq. 10a to ${}^\alpha N_t^o = \Delta^\alpha N_t^m$ at time, t , and ${}^\alpha N_t^o = \Sigma(\Delta^\alpha N_t^{u(i)} + \Delta^\alpha N_t^{l(i)})$ for all subsequent time. The redistribution of species α back from the orogene, as defined by eq. 10b, remains as before, but we only consider atoms of age t (the model is run in the usual way prior to time, t , for the purpose of having the correct value of $\Delta^\alpha N_t^m$). We can now count the total atoms of species α and age t residing in the final, present-day model configuration of the upper and lower crust. A similar inventory is taken for each age of initial appearance in the crust, and the mean age is then calculated in the usual manner:

$$\bar{T}_\alpha = \int_t \frac{\sum_i ({}^\alpha N_t^{u(i)} + {}^\alpha N_t^{l(i)}) \cdot t}{\sum_i ({}^\alpha N_t^{u(i)} + {}^\alpha N_t^{l(i)})} \quad (25)$$

where the summation for $i = 1$ to 11 is made today for the final configuration of reservoirs, and the integration is for all age t of species α . For version I of the plumbotectonics model $\bar{T}_{204\text{Pb}} = 3.11$ b.y., $\bar{T}_{238\text{U}} = 3.15$ b.y., and $\bar{T}_{232\text{Th}} = 3.23$ b.y. Although these mean ages may at first seem surprisingly old, the strong skewness toward an early mantle participation in the orogene and the preference of these elements toward enrichment in crustal reservoirs will, indeed, force an early separation of Pb, U, and Th from the mantle. By comparison, if a constant value of $(\Delta M_t^m / M_t^m) = 1/32$, which maintains a rather similar set of final concentration values, is chosen to reflect uniform mantle participation in the orogene, we can significantly lower the mean age to $\bar{T}_{204\text{Pb}} = 2.09$ b.y., $\bar{T}_{238\text{U}} = 2.10$ b.y., and $\bar{T}_{232\text{Th}} = 2.18$ b.y. Obviously, such a change in $(\Delta M_t^m / M_t^m)$ will also impact the isotopic compositions generated by the model, and in the following section we will put forth an alternate version of plumbotectonics that attempts to best fit all constraints.

VERSION II OF PLUMBOTECTONICS

Since the original version of the plumbotectonics model was specified several years ago, we have had an opportunity to add substantially to our knowledge of the lead-isotopic character of the Earth. Likewise, other information serving to advance our understanding of mantle and crust evolution has come from intensive studies of a wide range of rocks spanning much of geologic time. Therefore, reexamination of the plumbotectonics model seems appropriate with an eye toward adjusting some of the model parameters and formulations to bring them into closer agreement with observational data. This section of the paper is devoted to presenting an alternate version (version II) of plumbotectonics, which achieves that purpose with only slight modification of the original model equations. In order to avoid the circuitous reasoning that can easily accompany the use of dubious external constraints, primary emphasis here is given to reconciling the model with current rock and ore lead-isotope data themselves rather than with other chemical and physical criteria. We only ask that our model not be in obvious violation of known external constraints; then proceed to make a best fit to the lead-isotope systematics.

The most powerful control on our model is exerted by the fields defining the present-day lead-isotopic composition for major reservoirs as shown in Fig. 1. We have the greatest amount of data and the best documentation of age and geologic environment for these young rocks and ores. Growth curves that attempt to trace lead-isotope evolution as a function of sample age become more vague with increasing geologic time, and considerable latitude exists in choosing model parameters to match such curves. What information we do have on critically establishing the time and extent of reservoir interaction, therefore, is mainly provided by the relative positioning of the modern lead-isotope fields. Repeated computations with our model have shown a consistent relationship between the lead-isotopic composition of modern mantle, upper crustal, lower crustal, and orogene reservoirs and the mean age of lead in the crust. The slope of a line drawn through the nearly linear array of these points on a $^{207}\text{Pb}/^{204}\text{Pb}$ vs. $^{206}\text{Pb}/^{204}\text{Pb}$ diagram for a wide range in model parameters can be translated approximately into a mean age as would be calculated from eq. 25; i.e., m (slope) = $\Delta^{207}\text{Pb}/\Delta^{206}\text{Pb} \simeq (1/137.88) \cdot (e^{\lambda^{235}\bar{T}} - 1)/(e^{\lambda^{238}\bar{T}} - 1)$. For example, version I of plumbotectonics, which has been shown to have a $T_{204\text{Pb}} = 3.11$ b.y., yields a "slope" age of 2.96 b.y. ($m = 0.217$). Changes in model parameters that keep $\bar{T}_{204\text{Pb}}$ constant while affecting communication among the reservoirs influence the spread in final isotopic composition of the various reservoirs but do little to alter the slope.

In comparing the *average* fields for modern mantle, upper crustal, lower crustal, and orogene lead-isotopic compositions as drawn in Fig. 1 with the model computed isotopic compositions, version I of plumbotectonics (symbol +) can no longer be held as a good fit to the observational data. In partic-

ular, the model values for mantle and lower crustal lead lie outside the current estimates for their *average* fields. The discrepancy is seen to lie in a direction that oversteepens the slope, m , on the $^{207}\text{Pb}/^{204}\text{Pb}$ vs. $^{206}\text{Pb}/^{204}\text{Pb}$ diagram and, correspondingly, overestimates the mean age of lead in the crust. From a cursory consideration of the model parameters, several factors are seen to bear on the mean age and might be adjusted to produce a more satisfactory fit between model and data. Included among them are: (1) the timing of the orogenies; (2) the mantle participation function, $(\Delta M_t^m/M_t^m)$; (3) the mass of the newly formed segments of upper and lower crust, $M_t^{u(g)}$ and $M_t^{l(g)}$; (4) the erosional, f_v , and areal, f_h , functions applied to preexisting crust; and (5) the several isotope distribution functions involved in creating and dispersing the orogenes.

Careful thought was given to our initial modelling of the build-up and destruction of continental crust as specified by factors (3), (4), and (5). A drastic change in this aspect of plumbotectonics would be required in order to effect the desired shift in final isotopic composition, and we choose not to attack the problem from this direction. Whether orogenic activity peaked at some intermediate time(s), such as 2.7 b.y. ago or 1.8 b.y. ago, however, remains a real possibility that must eventually be resolved. There is little doubt that version I does overestimate the antiquity of crustal differentiation — an early but largely destroyed temporary crust would be moot to the model — but the most likely culprit to have produced this bias is the strong skewness of the mantle participation function toward the first few orogenies. Consequently, we have experimented with a number of alternate numerical evaluations of $(\Delta M_t^m/M_t^m)$, which range from that of version I to that of constant participation in all orogenies. An example of the latter situation, for which $(\Delta M_t^m/M_t^m) = 1/32$ at all times, has already been mentioned as yielding a $\bar{T}_{204\text{Pb}}$ of 2.09 b.y. Clearly, only by significantly reducing the skewness of the mantle participation function can we obtain a mean age of lead in the crust in reasonable agreement with the observational data. Many variants of the mantle participation function that would rectify the oversteepened slope of the model's modern lead-isotopic composition in Fig. 1 are possible as well as geologically plausible. Our new choice largely avoids the fine-structure weighing of individual orogenies, yet recognizes a broad peaking of such activity at intermediate ages. Because of the interdependence of all model parameters on each other, the adjustment in the mantle participation function does necessitate a slight change in the lead partitioning ratios, too, in order to make a good match with the *average* reservoir fields.

Another way to reduce the mean age of lead in the crust would have been by changing the timing of the orogenies. In particular, the small volume of rocks known to be older than 3.6 b.y. and the virtual absence of rocks older than 3.8 b.y. suggests the elimination of the first orogeny at 4.0 b.y. ago from our model. However, some old rocks do contain lead-isotopic evidence of a ~4.0 b.y. differentiation event, and we propose as a compromise to retain all the eleven orogenies of the original model but to reflect less activity

TABLE V

Parameters of the plumbotectonics lead isotope evolution model (version II)

- I. Starting conditions (all material in undifferentiated upper mantle to 500 km depth [800×10^{24} g] at 4.0 b.y. ago):
- A. Lead isotopic composition: $^{206}\text{Pb}/^{204}\text{Pb} = 10.32$; $^{207}\text{Pb}/^{204}\text{Pb} = 12.12$; $^{208}\text{Pb}/^{204}\text{Pb} = 30.56$
- B. Element abundances: $^{204}\text{Pb} = 37.0 \times 10^{15}$ moles; $^{238}\text{U} = 344 \times 10^{15}$ moles; $^{232}\text{Th} = 1310 \times 10^{15}$ moles
- II. Eleven orogenies evenly spaced in time; i.e., at 4.0 b.y., 3.6 b.y., 3.2 b.y., ..., 0.0 b.y. During each orogeny:
- A. Of the material entering the orogene, (1) the mantle contributes 1/64 of itself at 4.0 b.y., 1/32 at 3.6 b.y., 1/16 at 3.2 b.y. and 2.8 b.y., 1/32 at 2.4 b.y., 2.0 b.y., and 1.6 b.y., and 1/64 thereafter; (2) the upper crust contributes 3/10 of remaining amounts of each older segment (erosional function); and (3) the total crust contributes 1/10 of remaining amounts of each older segment (areal function). Components (2) and (3) do not pertain to the first orogeny.
- B. The contents of the orogene are chemically and isotopically homogenized.
- C. 2.6×10^{24} g of new upper crust and 2.6×10^{24} g of new lower crust are created, and remaining material returns from the orogene to the mantle. Lead, uranium, and thorium distribute themselves according to the following partitioning ratios:

Element	Mantle	Upper crust	Lower crust
Lead	0.036	0.742	0.222
Uranium	0.024	0.852	0.124
Thorium	0.020	0.786	0.194

III. Ending conditions (prevailing today at time of final orogeny):

A. Lead isotopic composition:

Environment	$\frac{^{206}\text{Pb}}{^{204}\text{Pb}}$	$\frac{^{207}\text{Pb}}{^{204}\text{Pb}}$	$\frac{^{208}\text{Pb}}{^{204}\text{Pb}}$	$\frac{^{238}\text{U}}{^{204}\text{Pb}}$	$\frac{^{232}\text{Th}}{^{238}\text{U}}$
Mantle	18.01	15.47	37.71	8.35	3.52
Orogene	18.89	15.61	38.81	11.32	3.62
Upper crust contributed to orogene	19.44	15.70	39.19	13.23	3.43
Lower crust contributed to orogene	17.54	15.39	38.67	6.29	5.91

B. Element abundances and concentrations:

Reservoir	Mass ($\times 10^{24}$ g)	Abundances ($\times 10^{15}$ moles)			Average concentration (ppm)		
		^{204}Pb	^{238}U	^{232}Th	Pb	U	Th
Mantle	775.2	19.0	158	555	0.38	0.050	0.17
Upper crust	7.0	10.3	137	467	23.5	4.7	15.5
Lower crust	17.8	7.7	49	288	6.4	0.65	3.7

TABLE VI
Lead-isotopic composition for reservoir growth curves generated by the plumbotectonics model (version II)

<i>t</i> (b.y.)	Mantle				Orogene				Upper crust				Lower crust			
	<u>206Pb</u>		<u>208Pb</u>		<u>206Pb</u>		<u>208Pb</u>		<u>206Pb</u>		<u>208Pb</u>		<u>206Pb</u>		<u>208Pb</u>	
	204Pb	207Pb	204Pb	207Pb	204Pb	207Pb	204Pb	207Pb	204Pb	207Pb	204Pb	207Pb	204Pb	207Pb	204Pb	207Pb
4.0	10.32	12.12	30.56	10.32	10.32	12.12	30.56	10.32	10.32	12.12	30.56	10.32	10.32	12.12	30.56	10.32
3.6	11.36	13.25	31.40	11.38	11.38	13.26	31.41	11.54	11.54	13.44	31.48	10.91	10.91	12.76	31.32	10.91
3.2	12.32	14.00	32.22	12.36	12.36	14.03	32.24	12.62	12.62	14.25	32.38	11.81	11.81	13.55	32.16	11.81
2.8	13.21	14.50	32.99	13.31	13.31	14.56	33.07	13.65	13.65	14.78	33.30	12.67	12.67	14.12	33.02	12.67
2.4	14.03	14.83	33.72	14.29	14.29	14.96	33.96	14.63	14.63	15.13	34.19	13.46	13.46	14.51	33.86	13.46
2.0	14.80	15.05	34.42	15.17	15.17	15.20	34.78	15.59	15.59	15.37	35.09	14.21	14.21	14.78	34.69	14.21
1.6	15.53	15.21	35.11	15.98	15.98	15.36	35.58	16.46	16.46	15.52	35.94	14.93	14.93	14.98	35.51	14.93
1.2	16.21	15.32	35.79	16.86	16.86	15.49	36.49	17.27	17.27	15.60	36.76	15.62	15.62	15.13	36.31	15.62
0.8	16.85	15.39	36.44	17.59	17.59	15.55	37.29	18.06	18.06	15.66	37.60	16.29	16.29	15.24	37.11	16.29
0.4	17.45	15.44	37.08	18.27	18.27	15.59	38.06	18.78	18.78	15.69	38.41	16.93	16.93	15.32	37.90	16.93
0.0	18.01	15.47	37.71	18.89	18.89	15.61	38.81	19.44	19.44	15.70	39.19	17.54	17.54	15.39	38.67	17.54

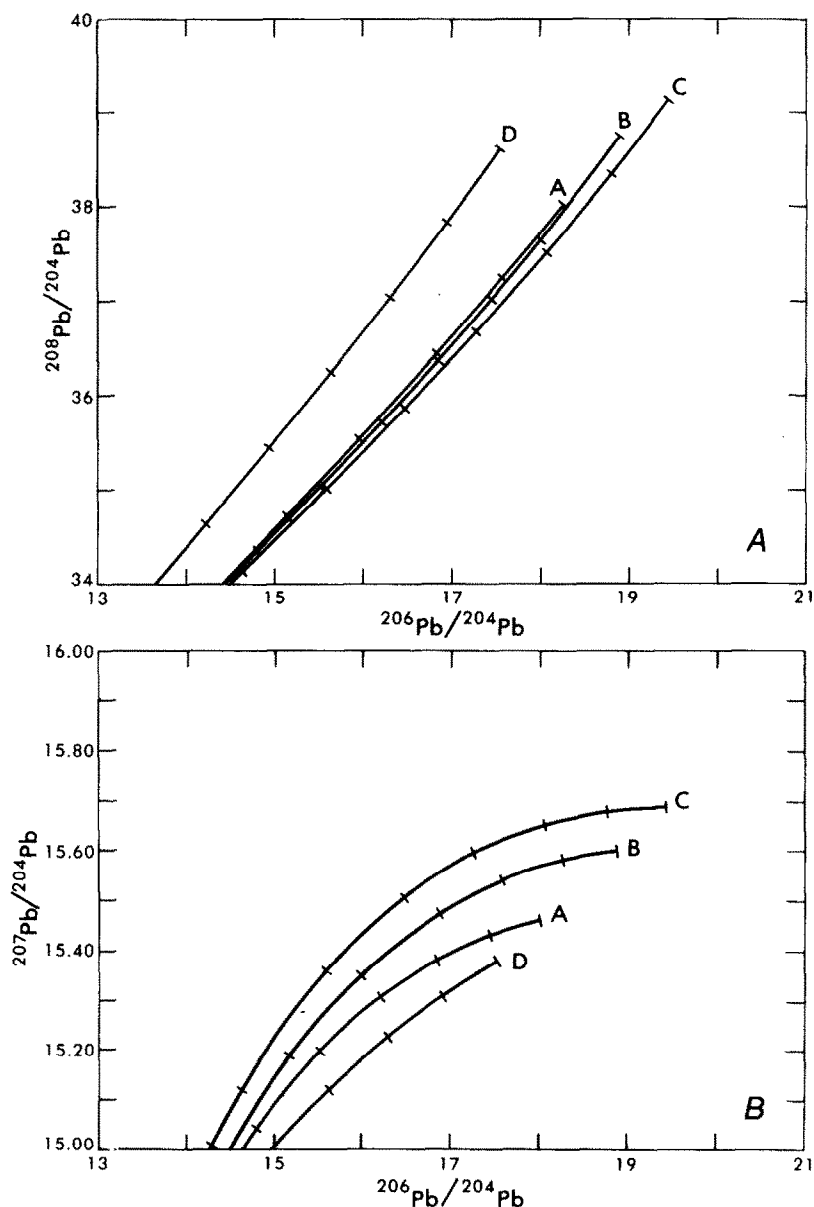


Fig. 5. A, B. Lead-isotope evolution curves generated by the plumbotectonics model (version II) for the mantle (A), orogene (B), upper crust contributed to the orogene (C), and lower crust contributed to the orogene (D). The labelled end points represent the modern isotopic composition for these environments and may be compared with the model representation in Fig. 2. Tick marks along each curve indicate progressively older time in 0.4-b.y. increments.

by reducing ($\Delta M_T^T / M_T^T$) for the earliest orogenies.

From the aforementioned considerations we determine an updated set of model parameters for version II of plumbotectonics as given in Table V. From these parameters we can calculate the isotope composition and concentration values presented in Table VI and Figs. 5, 6, and 7. Also, we have $T_{204\text{Pb}} = 2.29$ b.y., $\bar{T}_{238\text{U}} = 2.39$ b.y., and $\bar{T}_{232\text{Th}} = 2.46$ b.y., which are considerably reduced over those mean ages obtained for version I of the plumbotectonics model.

Obviously, even this rather complex model is an oversimplification of actual geologic conditions for the Earth. Each tectonic domain undergoing orogenic development has unique source-rock conditions, geometry, duration, and subsequent erosional history, which will influence its detailed lead-isotope patterns. In a general way, however, attempts at modelling isotopic evolution do provide important insights into and constraints on the problem of gross mantle and crust interaction. We find the model lead-evolution curves for the mantle, lower crust, upper crust, and orogene to be useful references in interpreting rock and ore isotope data acquired from a wide variety of regional and topical studies. Also, by demonstrating the feasibility of a mixing model that generates isotopic compositions closely matching those observed for major geologic environments, a major contribution is made toward resolving the controversy over a basically static versus dynamic Earth. Whether individual mantle and crustal reservoirs have behaved as

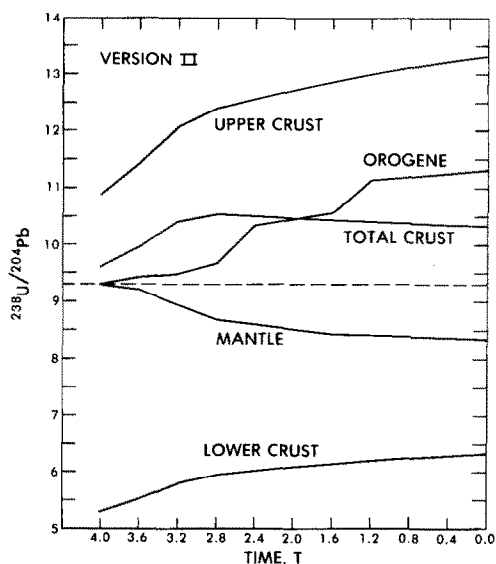


Fig. 6. $^{238}\text{U}/^{204}\text{Pb}$ generated by the plumbotectonics model (version II) for the mantle, orogene, and newly formed segments of upper crust, lower crust, and total crust as a result of each orogeny. The horizontal dashed line represents the entire system, which has a constant value of 9.30.

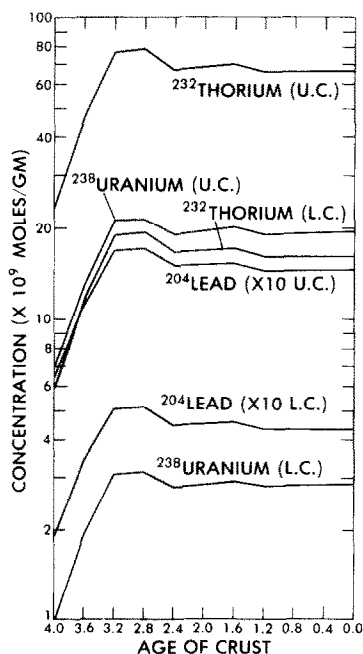


Fig. 7. ^{204}Pb , ^{238}U , and ^{232}Th concentrations in individual segments of upper (u.c.) and lower (l.c.) crust generated by the plumbotectonics model (version II). The lower values of the first two orogenies reflect the early, reduced differentiation of the mantle given by this version of the model; otherwise, there is no significant change in concentration with time.

essentially isolated chemical systems for much of geologic time or whether they exist in secular balance involving significant transfer of material among reservoirs is fundamental to our understanding of the planet's differentiation history.

THE ROLE OF THE MANTLE IN PLUMBOTECTONICS

The plumbotectonics model was primarily derived to explain the complementary partitioning of Pb, U, and Th between newly formed segments of upper and lower continental crust. In this context, the mantle could be represented as one uniform reservoir, which, furthermore, need not have a very different U-Pb ratio than the total crust. From direct observation of MORB and various oceanic island volcanic rocks, however, we know that considerable isotopic variability does exist in lead coming from the mantle (Figs. 1 and 8). Apparently, the scale of mixing accomplished in an orogene averages over these isotopic heterogeneities and destroys much of their separate identity. If the mantle contribution to the orogene is dominated by subducted oceanic crust, we might expect it to consist largely of a MORB

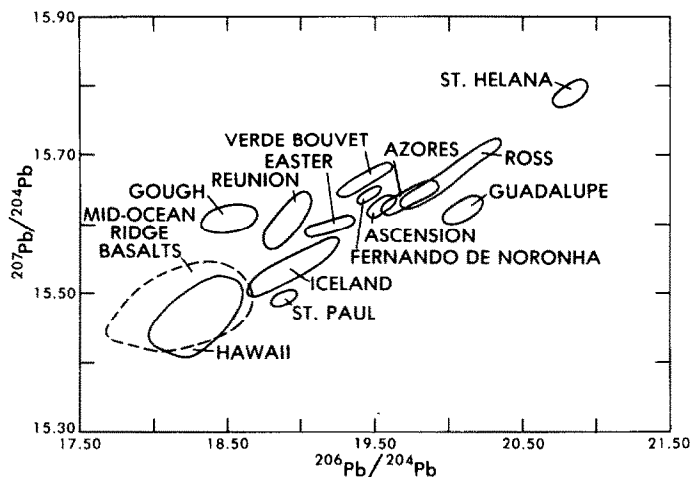


Fig. 8. $^{207}\text{Pb}/^{204}\text{Pb}$ vs. $^{206}\text{Pb}/^{204}\text{Pb}$ diagram for mid-ocean ridge basalts and oceanic island volcanic rocks. Data taken from Tatsumoto (1978) and Sun (1980).

component with only minor amounts added from other sources. In fact, for a wide range in reasonable parameter values, the model almost demands a mantle isotopic composition lying in or near the MORB field on a $^{207}\text{Pb}/^{204}\text{Pb}$ vs. $^{206}\text{Pb}/^{204}\text{Pb}$ diagram, and we see little influence of the more radiogenic oceanic island lead. Either the source region giving rise to MORB makes up the greatest volume of the mantle, or else other parts of the mantle with different lead-isotopic characteristics are effectively prevented from contributing to the orogenic budget.

We should emphasize here that according to plumbotectonics, island-arc volcanic rocks, which might be related to another mantle reservoir distinct from the MORB source, represent mixtures of MORB and continental-derived material brought together in the orogenic environment. According to this scheme, primitive arcs far removed from continents closely resemble MORB in lead isotopic composition and more mature arcs contain, in addition, significant proportions of lead from adjacent continents. Combined Pb, Nd, and Sr isotopic analyses on the same suites of island-arc rocks are as yet too few to firmly establish this mixing model, but where comparison can be made such an interpretation is certainly possible (as, for example, in the Sierra Nevada; Doe and Delevaux, 1973; Kistler and Peterman, 1978; DePaulo, 1980).

Another major mantle reservoir — this one serving as the source of continental flood basalts (CFB) — has been proposed on the basis of a Nd and Sr isotopic composition quite different from that producing MORB (Wasserburg and DePaulo, 1979; Jacobsen and Wasserburg, 1979). Because CFB show isotopic patterns suggestive of a simple, closed-system evolution in a

"chondritic" mantle prior to eruption, a model has been inferred whereby undepleted primitive mantle is batch-processed to yield continental crust and a residual depleted mantle. Accordingly, CFB are thought to represent direct access to the deep primitive mantle, and MORB are thought to be a sampling of the shallower residual mantle. Oceanic island volcanic rocks, which show a spread in Nd and Sr isotopic composition between CFB and MORB, are interpreted as primitive mantle magmas that were contaminated during ascent through the upper residual mantle. Unfortunately, most CFB appear to have been strongly influenced in their lead isotopes by the thick continental crust (and immediately underlying mantle?) that was penetrated to reach the surface. Studies of plateau and other basalts from continental areas reveal a wide range in lead-isotopic composition with both excesses and deficiencies in radiogenic isotopes suggestive of upper and lower crustal contamination, respectively. The least-contaminated basalts give some indication of convergence on a lead isotopic composition approximating that of the orogene, but little clue is offered to distinguish between an actual orogenic or a primitive mantle source for the lead.

Much attention in the use of lead isotopes to understand the character of the mantle has been given to oceanic island volcanic rocks (Oversby and Gast, 1970; Tatsumoto, 1978; Sun, 1980). Their unique positions on $^{207}\text{Pb}/^{204}\text{Pb}$ vs. $^{206}\text{Pb}/^{204}\text{Pb}$ and $^{208}\text{Pb}/^{204}\text{Pb}$ vs. $^{206}\text{Pb}/^{204}\text{Pb}$ diagrams require a separate origin from the other terrestrial reservoirs shown in Fig. 1. If oceanic island volcanic rocks do represent a mixture derived from undepleted and depleted mantle sources as proposed by Wasserburg and DePaolo (1979), we face the dilemma of having a most peculiar radiogenic lead for the undepleted source. While the unradiogenic end member of such a mixing line readily identifies itself with the depleted MORB source, support for associating the radiogenic end member with the undepleted CFB source is virtually absent. Some intermediate position near to that of orogenic lead, which might better qualify for such a primitive "chondritic" reservoir, would leave unexplained the more radiogenic oceanic island samples.

Our preferred interpretation of the lead-isotopic field defined by oceanic island volcanic rocks calls for a mantle heterogeneity established much later than the early differentiation of the Earth. The shallow slope of this array on a $^{207}\text{Pb}/^{204}\text{Pb}$ vs. $^{206}\text{Pb}/^{204}\text{Pb}$ diagram seen in Fig. 8 is particularly suggestive of an isolation of separate sources for individual islands with mean lifetimes of ~ 1.5 b.y. We are not certain that any important worldwide mantle differentiation event need be attached to this "age". Rather, we suggest that the "age" represent an average residence time for isolated pods of anomalous mantle that eventually produce the oceanic island volcanic rocks. These pods could be eclogite bodies formed by the subduction of oceanic crust that was perhaps modified through involvement in arc magmatism. As such, they would exist as plums in a pudding until some subsequent thermal perturbation caused their selective partial or complete melting. In order to satisfy the lead-isotopic characteristics observed in oceanic island volcanic rocks, the

Pods must derive from a source material of isotopic composition similar to that giving rise to MORB, but with a significant increase in $^{238}\text{U}/^{204}\text{Pb}$ and $^{232}\text{Th}/^{204}\text{Pb}$ ratios. Pods with short lifetimes (0.5 b.y.) and (or) small U/Pb and Th/Pb enrichments will not depart significantly from the MORB isotopic field; pods with long lifetimes (1–2 b.y.) and (or) large U-Pb and Th/Pb enrichments will have more radiogenic isotopic compositions. Recently, Chase (1980), in a similar approach, has calculated such a range of residence times based on $^{206}\text{Pb}/^{204}\text{Pb}$ vs. $^{207}\text{Pb}/^{204}\text{Pb}$ linear arrays from several individual oceanic islands. The proximity of oceanic islands to mid-ocean ridges and mantle hot-spots hints that melting of the pods takes place primarily in these areas of high heat flow. Considering the relatively rapid rate of upper mantle convection implicit in plate tectonics, it does not seem unreasonable that pods such as we envision here will generally not survive for more than 2 b.y. Again, we repeat that oceanic island volcanic rocks give no evidence in their lead isotopes of being derived from a primitive source material or other mantle reservoir created early in the Earth's history.

This brief discussion of the role of the mantle in plumbotectonics puts forth some of our speculations on the lead isotope fine-structure of this reservoir. If the average mantle and the *total* crust contain lead evolved with a quite similar U-Pb ratio, this behavior would be in marked contrast to that of the Rb-Sr and Sm-Nd systems. Consequently, we lose much of the isotopic leverage provided by a highly fractionated parent-daughter system in determining how large a fraction of the mantle does communicate with the crust. Neither are we provided with a clear distinction between an undepleted primitive mantle and a residual depleted mantle. The similarity in effective U-Pb ratio, however, may only partially be due to actual coherence of uranium and lead during the differentiation process. At least as important a factor may be a significant bidirectional exchange of lead among all reservoirs via the orogene — a process that tends to further reduce isotopic differences by recycling previously isolated systems. In future papers we hope to explore the U-Th-Pb systematics in greater detail, and to relate plumbotectonics to other current models of lead, strontium, and neodymium isotopes.

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